Project Proposal

NexaHR — AI-Powered Employee & HR Policy Copilot

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Project Domain: Enterprise AI, Agentic Systems, Retrieval-Augmented Generation (RAG), Full-Stack Web Development

Abstract

NexaHR is a full-stack enterprise web application designed to automate and streamline employee HR inquiries using an Agentic RAG architecture. Enterprise HR departments maintain extensive documentation covering leave, payroll, benefits, and conduct, leading to high inquiry volumes and repetitive administrative workloads. NexaHR addresses this by deploying specialized AI agents (Query, Retrieval, Policy Reasoning, Action, and Response) to deliver grounded, context-aware answers with verifiable source citations, automated form assistance, policy versioning, and role-based access control.

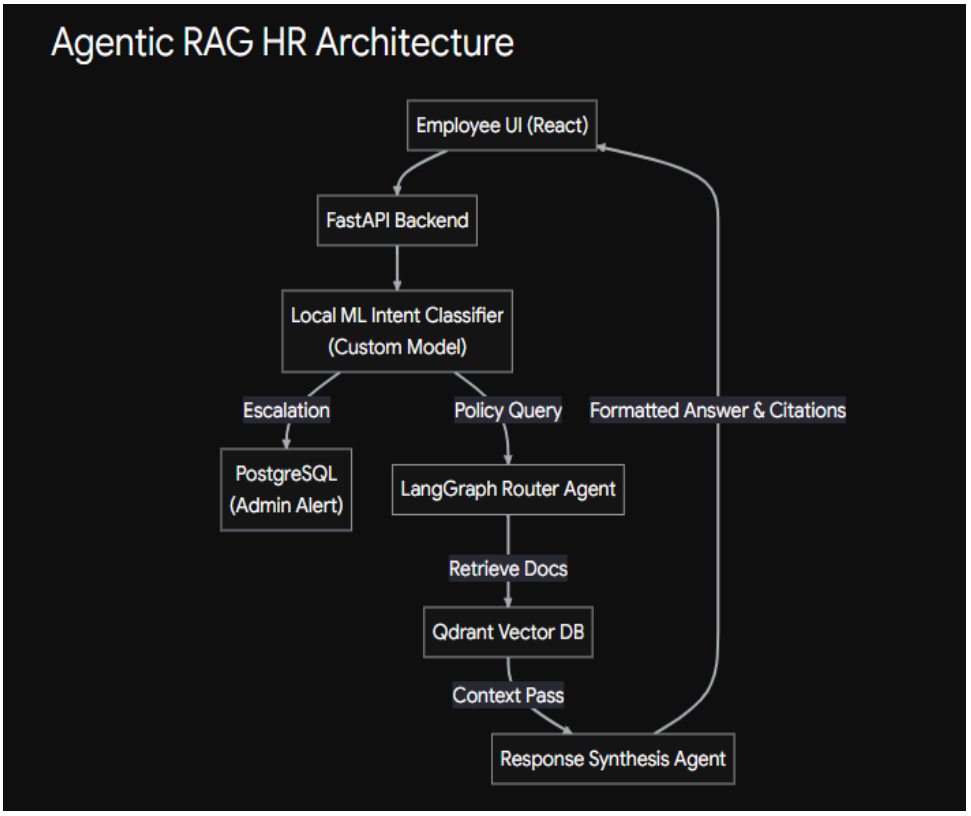
1. Project Objective

- Develop a full-stack, enterprise-grade web application that acts as an HR Copilot.
- Automate the retrieval and explanation of HR policies for employees using natural language.
- Implement a hybrid AI system (Custom Machine Learning + Agentic RAG) to ensure dynamic, real-time query resolution.
- Deploying on Cloud with Devops Setup and keeping project Opensource.

2. Real-World Application

- **Enterprise HR Support:** Allows employees to self-serve policy questions (e.g., leave calculations, remote work eligibility) instantly with verified citations.
- **Onboarding Automation:** Helps new hires interactively navigate company wikis and codes of conduct, reducing the administrative burden on human HR staff.

3. System Architecture Flow



4. Tech Stack & Implementation Details

- **Frontend:** React + Tailwind CSS (Fully integrated user interface).
- **Backend:** Python + FastAPI (RESTful API architecture).
- **Deployment:** Azure, Docker, Github Actions
- **Databases:**
    - PostgreSQL (Relational DB for users, chat logs, and state management).
    - Qdrant (Vector DB for storing semantic embeddings of policies).
- **Machine Learning & Agentic Layer:**
    - **ML Model:** A custom-trained NLP classification model to analyze user intent *before* LLM routing.
    - **Generative AI:** LangGraph for multi-agent workflows and LangChain for RAG.

5. Project Scope Compliance (Guideline Verification)

- **Application-Based:** YES. It is a functioning Enterprise SaaS web application.
- **Not Dataset-Based:** YES. Operates dynamically on live inputs and live document ingestion, not static CSV files.
- **Custom Database:** YES. Utilizes an independently designed relational PostgreSQL schema and Qdrant vector store.
- **Integrated Interface:** YES. Features a complete React frontend connected seamlessly to the backend API.

- **ML Implemented:** YES. Integrates a custom-trained machine learning classification model to route queries, satisfying the requirement for implemented ML beyond just API wrappers.